

Homework 1 - W17 - Solutions

- **Due date:** Saturday, February 18, 2017 at **midnight** (11:59:59 pm)
- Partial or unsatisfactory solutions will only receive a fraction of the points.

Electronic Submission Guidelines

- All homework submissions must be electronic and submitted to the corresponding EEE dropbox.
- **Only PDF files are accepted.** Please **DO NOT** submit in any other format including plain text files, pictures, Microsoft Word documents, L^AT_EX source files, compressed archives, etc.
- Please submit **only one file** for solutions to all problems.
- PDF files must be **typed**. Please **DO NOT** submit or include handwritten/drawn parts (including those from handwriting/drafting applications) or anything from a camera/scanner picture.
- Please name your submission "<student ID number>.pdf", e.g. "12345678.pdf", and indicate your full name and student ID number in the file.

P1 (15 Points) Based on the textbook (**Kurose-Ross**) problem **Chapter 1, P2** on page 71. Equation 1.1 on the book gives a formula for the end-to-end delay of sending one packet of length L over N links of transmission rate R , which is

$$d = N \cdot \frac{L}{R} \quad (1)$$

- Assume one of the N links degraded and its transmission rate becomes $R' < R$. Find the end-to-end delay of the packet.
- Suppose the degraded link is the first one of the N links. Generalize the end-to-end delay for sending P such packets back-to-back over the N links.
- Suppose the degraded link is the last (N -th) one. Generalize the end-to-end delay as in (b).

Solution

- (a) **(5 pt)** The given formula only counts transmission delay. Therefore, the end-to-end delay is the **total transmission time** over $(N - 1)$ normal links and one degraded link, which is

$$d = (N - 1) \cdot \frac{L}{R} + \frac{L}{R'}$$

- (b) **(5 pt)** The end-to-end delay of the first packet is the expression we got in (a). Each other packet is L/R' time behind the previous one. Hence, the answer is

$$d = (N - 1) \cdot \frac{L}{R} + P \cdot \frac{L}{R'}$$

- (c) **(5 pt)** The same with (b).

P2 (10 Points) Solve the textbook (**Kurose-Ross**) problem **Chapter 1, P8** on page 72. (Suppose users share a 3 Mbps link...)

Solution

- (2 pt)** When circuit switching is used, each user needs a dedicated link of 150 kbps bandwidth. Therefore, the total number of supported users is $3 \times 10^6 / 150 \times 10^3 = 20$.
- (2 pt)** $1/10 = 0.1$.
- (3 pt)** The answer is the product of (1) the probability that n specific users are transmitting, (2) the probability that the other users are not transmitting, and (3) the number of combinations to choose such n users from the total of 120 users, which is

$$0.1^n \cdot (1 - 0.1)^{120-n} \cdot C_n^{120},$$

where C_k^n is the combination number (binomial coefficient), i.e.

$$C_k^n = \binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

- (d) (3 pt) There are 121 different possible cases in total, where there are *exactly* 0, 1, 2, ..., 120 users transmitting simultaneously, respectively. Hence, the answer is the sum of probability from multiple cases, which is

$$\sum_{n=21}^{120} 0.1^n \cdot (1 - 0.1)^{120-n} \cdot C_n^{120},$$

or equivalently

$$1 - \sum_{n=0}^{20} 0.1^n \cdot (1 - 0.1)^{120-n} \cdot C_n^{120}$$

The numerical value is approximately **0.007941**.

P3 (10 Points) Solve the textbook (**Kurose-Ross**) problem **Chapter 1, P24** on page 75. (Suppose you would like to urgently deliver 40 terabytes data from Boston to Los Angeles...)

Solution

The transmission time using the dedicated link is $40 \times 10^{12} \times 8 / (100 \times 10^6) = 3.2 \times 10^6$ sec ≈ 37.04 d. Hence, it takes approximately 37 days to transmit these data via this dedicated link, while FedEx overnight delivery could deliver these data in approximately one day. Since it is urgent, we would prefer **FedEx overnight delivery**.

P4 (10 Points) Solve the textbook (**Kurose-Ross**) problem **Chapter 1, P25** on page 76. (Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected...)

Solution

- (a) (2 pt) The propagation delay is $d_{\text{prop}} = 20000 \times 10^3 / (2.5 \times 10^8) = 0.08$ sec. Therefore, the bandwidth-delay product is

$$R \cdot d_{\text{prop}} = 2 \times 10^6 \cdot 0.08 = 1.6 \times 10^5 \text{ bit} = \mathbf{160 \text{ kbit}}$$

- (b) (2 pt) The same with (a).
 (c) (1 pt) The bandwidth-delay product is the maximum number of bits that could be in the link.
 (d) (2 pt) The width of a bit is $20000 \times 10^3 / (1.6 \times 10^5) = \mathbf{125 \text{ m}}$. It is **longer** than a football field.
 (e) (3 pt) The general expression for the width of a bit is

$$m / \left(R \cdot \frac{m}{s} \right) = \frac{s}{R}$$

P5 (20 Points) Consider the network shown in Fig. 1. Routers 1, 2, 3, and 4 forward packets at fixed rates $R_1=0.5$ Mbit/s, $R_2=1.0$ Mbit/s, $R_3=0.5$ Mbit/s, and $R_4=2.0$ Mbit/s, respectively. Routers 1 and 2 start the transmission of packets A and B simultaneously. Router 2 transmits packets B and C back-to-back. The buffers of routers 3 and 4 are empty. The size of packets A, B, and C is $L=1$ Mbit. **Assume processing and propagation delay are negligible.** (*Hint: Routers transmit packets at fixed deterministic rates.*)

- (a) Find the end-to-end delay of packets A, B, and C.
 (b) Solve the same problem, when $R_3=2.0$ Mbit/s, and the rates of other routers remain unchanged.
 (c) Find the end-to-end delay of packet C, when $R_3=2.0$ Mbit/s, $R_4=1.0$ Mbit/s, and the rates of other routers remain unchanged.

Solution

- (a) (7 pt) Compute the timelines of routers as follows:

Router 1 (0.5 Mbps) 0.0 – 2.0 sec Packet A

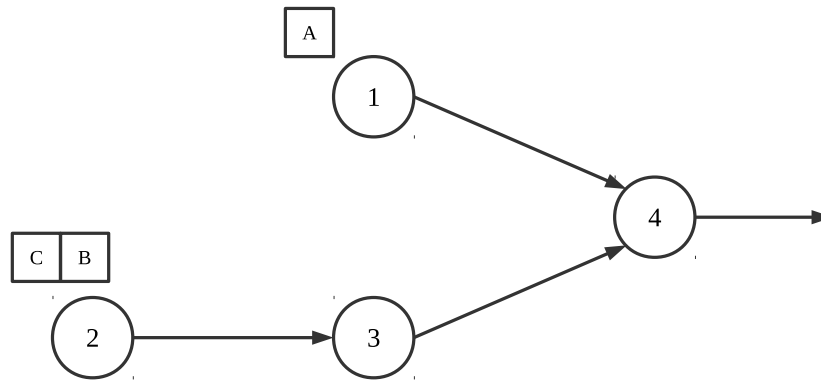


Figure 1. Problem 1

Router 2 (1.0 Mbps)	0.0 – 1.0 sec	Packet B
	1.0 – 2.0 sec	Packet C
Router 3 (0.5 Mbps)	1.0 – 3.0 sec	Packet B
	3.0 – 5.0 sec	Packet C
Router 4 (2.0 Mbps)	2.0 – 2.5 sec	Packet A
	3.0 – 3.5 sec	Packet B
	5.0 – 5.5 sec	Packet C

Hence, the end-to-end delay of packets A, B, and C is respectively **2.5 sec, 3.5 sec, 5.5 sec**.

(b) (7 pt) Compute the timelines of routers as follows:

Router 3 (2.0 Mbps)	1.0 – 1.5 sec	Packet B
	2.0 – 2.5 sec	Packet C
Router 4 (2.0 Mbps)	1.5 – 2.0 sec	Packet B
	2.0 – 2.5 sec	Packet A
	2.5 – 3.0 sec	Packet C

Hence, the end-to-end delay of packets A, B, and C is respectively **2.5 sec, 2.0 sec, 3.0 sec**.

(c) (6 pt) Compute the timelines of routers as follows:

Router 4 (1.0 Mbps)	1.5 – 2.5 sec	Packet B
	2.5 – 3.5 sec	Packet A
	3.5 – 4.5 sec	Packet C

Hence, the end-to-end delay of packet C is **4.5 sec**.

P6 (10 Points) Consider a router whose transmission rate is R bit/s (fixed rate). The time between two consecutive packet arrivals in its (infinite) buffer is an exponentially distributed variable with rate λ . The length of the packets is an exponentially distributed variable with an average of L bits. At time 0, there is no packet in the buffer.

- Define T as the time when the last bit of the first packet is transmitted by the router. Find $\mathbf{E}[T]$.
- Suppose by time s sec, no packet has ever arrived. Find the probability that the first packet arrives before $(s + t)$ sec.
- Assume when packet A arrives, there are already N packets in the system, including the first one of them being served. Define d_{queue} as the queuing delay of packet A. Find $\mathbf{E}[d_{\text{queue}}]$.

Solution

- (a) (3 pt) T is the sum of (1) the **time to wait for the first packet** to come, and (2) the **time to transmit the first packet**, whose expectation is

$$\mathbf{E}[T] = \frac{1}{\lambda} + \frac{L}{R}$$

- (b) (3 pt) $P(X > s + t | X > s) = P(X > t) = e^{-\lambda t}$. Hence, $P(X < s + t | X > s) = 1 - e^{-\lambda t}$.

- (c) (4 pt) The queuing delay d_{queue} of packet A is the **transmission time** of the N packets that are already in the buffer when A arrives. Therefore,

$$\mathbf{E}[d_{\text{queue}}] = N \cdot \frac{L}{R}$$

P7 (15 Points) A router receives packets from N applications. The time between the generation of two consecutive packets of application $k, k \in 1, 2, \dots, N$ is an exponentially distributed variable with rate λ_k . Assume no packet is in the buffer at time 0.

- (a) Define T_1 as the inter arrival time of packets at the router. Find $\mathbf{E}[T_1]$.
 (b) Find the probability that the first packet received by the router is generated by application 1.
 (c) Define T_N as the earliest time when the router has received at least one packet from each of the N applications. For $\lambda_1 = \lambda_2 = \dots = \lambda_N = \lambda$ and $N = 3$, solve $\mathbf{E}[T_N]$

Solution

- (a) (4 pt) The inter arrival time T_1 is the **time to wait for the first packet** to come, whose expectation is

$$\mathbf{E}[T_1] = 1 / \sum_{j=1}^N \lambda_j$$

- (b) (4 pt) The probability that the first packet is generated by application 1 is

$$\lambda_1 / \sum_{j=1}^N \lambda_j$$

- (c) (7 pt) **Method 1 – Using time to next packet from a different application**, where

$$\mathbf{E}[T_3] = \frac{1}{3 \cdot \lambda} + \frac{1}{2 \cdot \lambda} + \frac{1}{\lambda} = \left(\frac{1}{3} + \frac{1}{2} + 1 \right) \cdot \frac{1}{\lambda} = \frac{11}{6} \cdot \frac{1}{\lambda}$$

Method 2 – Using expected number of packets multiplied by **average time to next packet**, where

$$\mathbf{E}[T_3] = \mathbf{E}[n_3] \cdot \frac{1}{3 \cdot \lambda} = \left(1 + \frac{3}{2} + 3 \right) \cdot \frac{1}{3 \cdot \lambda} = \frac{11}{2} \cdot \frac{1}{3 \cdot \lambda} = \frac{11}{6} \cdot \frac{1}{\lambda}$$

P8 (10 Points) A router receives packets from one application. The time between the generation of two consecutive packets of the application is an exponentially distributed variable with rate λ . The service time of the router is an exponentially distributed variable with rate μ . Assume no packet is in the buffer at time 0.

- (a) Find the probability that the first packet (packet A) finishes transmission before the second packet (packet B) arrives.
 (b) Define N as the number of packets in the buffer after packet B arrives. Find $\mathbf{E}[N]$.
 (c) Compute the average permanence time of packet B in the buffer (waiting time plus transmission time).
 (d) Define T as the time when the last bit of packet B is transmitted by the router. Find $\mathbf{E}[T]$.

Solution

- (a) (2 pt) There are two random events, (1) packet A finishes transmission with rate μ , and (2) packet B arrives with rate λ . Therefore, the probability that “A finishes transmission” happens first is

$$\frac{\mu}{\lambda + \mu}$$

- (b) (2 pt) **If packet A finishes transmission before packet B arrives**, there is one packet (just B itself) in the buffer after B arrives. Otherwise (i.e. **if packet B arrives before A finishes transmission**), there are two packets (both A and B) in the buffer. Hence,

$$\mathbf{E}[N] = \frac{\mu}{\lambda + \mu} \cdot 1 + \frac{\lambda}{\lambda + \mu} \cdot 2 = \frac{2 \cdot \lambda + \mu}{\lambda + \mu}$$

- (c) (2 pt) The permanence time of packet B is the transmission time of B plus the transmission time of any packet that is in the buffer when B arrives, which equals

$$\mathbf{E}[N] \cdot \frac{1}{\mu} = \frac{\mu}{\lambda + \mu} \cdot \frac{1}{\mu} + \frac{\lambda}{\lambda + \mu} \cdot \frac{2}{\mu} = \frac{1}{\lambda + \mu} + \frac{2 \cdot \lambda}{\lambda \cdot \mu + \mu^2}$$

- (d) (4 pt) **If packet A finishes transmission before packet B arrives**, then T is the sum of (1) the time to wait for A, (2) the time to transmit A, (3) the time to wait for B, and (4) the time to transmit B. **Otherwise (i.e. if packet B arrives before A finishes transmission)**, T is the sum of (1) the time to wait for A, (2) the time to transmit A, and (3) the time to transmit B. Hence,

$$\mathbf{E}[T] = \frac{\mu}{\lambda + \mu} \cdot \left(\frac{2}{\mu} + \frac{2}{\lambda} \right) + \frac{\lambda}{\lambda + \mu} \cdot \left(\frac{2}{\mu} + \frac{1}{\lambda} \right)$$